

T3 matched EFT Lagrangian terms: EFT_1_after_F_S1_S2

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F, S1, S2. Remaining active T3 fields: none.

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$F B_{\mu\nu}$
									$+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}\right)^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}\right)^2$ $+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}\right)^2}{6\epsilon}$ $+\frac{5}{6}\hbar\log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H1}\right)^2$ $+\frac{\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}\right)^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}\right)^2$ $+\frac{5\hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}\right)^2}{6\epsilon}$ $+\frac{5}{6}\hbar\log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \left(\lambda_{H2}\right)^2$		
								$-\frac{\hbar B_{\mu_1\mu_2}^2}{12\epsilon}$	$+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{4\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon}$ $+\frac{4\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{4\hbar\log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$ $-\frac{4\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$ $-\frac{4\hbar\log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}^1\lambda_{H1}^2}{\sqrt{3}\epsilon}$ $+\frac{\hbar\log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}^1\lambda_{H1}^2}{\sqrt{3}}$ $+\frac{\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}^1\lambda_{H2}^2}{\sqrt{3}\epsilon}$ $+\frac{\hbar\log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}^1\lambda_{H2}^2}{\sqrt{3}}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon}$ $-\frac{1}{24}\hbar\log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$ $-\frac{1}{24}\hbar\log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{24}\hbar\log\left(\frac{\mu_{S_1}^2}{M_{S_1}^2}\right)B_{\mu_1\mu_2}^2$ $-\frac{1}{24}\hbar\log\left(\frac{\mu_{S_2}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$F\,B_{\mu\nu}$	
									$\begin{aligned} &+\frac{hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}\right)^2}{2\epsilon}\\ &+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}\right)^2\\ &+\frac{5hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}\right)^2}{6\epsilon}\\ &+\frac{5}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H1}\right)^2\\ &+\frac{hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}\right)^2}{2\epsilon}\\ &+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}\right)^2\\ &+\frac{5hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}\right)^2}{6\epsilon}\\ &+\frac{5}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}\right)^2 \end{aligned}$	$\begin{aligned} &+C_{H2}H_{a_1}H_{a_1}^\dagger\\ &-H_{a_1}\mu^2H_{a_1}^\dagger\\ &+\frac{4h\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon}\\ &+\frac{4h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}\\ &+\frac{4h\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}\\ &-\frac{4h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}\\ &-\frac{4h\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\left(M_{S_1}^2-M_{S_2}^2\right)}\\ &+\frac{hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}\lambda_{H1}^2}{\sqrt{3}\epsilon}\\ &+\frac{h\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H1}\lambda_{H1}^2}{\sqrt{3}}\\ &+\frac{hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}\lambda_{H2}^2}{\sqrt{3}\epsilon}\\ &+\frac{h\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\lambda_{H2}\lambda_{H2}^2}{\sqrt{3}} \end{aligned}$	$\begin{aligned} &-\frac{13\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon}\\ &-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2\\ &-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \end{aligned}$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\begin{aligned} &-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)B_{\mu_1\mu_2}^2\\ &-\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2 \end{aligned}$				
									$\begin{aligned} &+\frac{hH_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger X_{H1,a_1,a_3}\lambda_{H1,a_2,a_4}\left(\lambda_{H1}\right)^2}{2\epsilon}\\ &+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger X_{H1,a_1,a_3}\lambda_{H1,a_2,a_4}\left(\lambda_{H1}\right)^2\\ &+\frac{3hH_{a_3}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}\right)^2}{2\epsilon}\\ &+\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger\left(\lambda_{H2}\right)^2 \end{aligned}$	$\begin{aligned} &+CH21H_{a_1}H_{a_1}^\dagger\\ &-H_{a_1}\mu^2H_{a_1}^\dagger\\ &+\frac{2h\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}\\ &+\frac{2h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\left(M_{S_1}^2-M_{S_2}^2\right)}\\ &+\frac{2h\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\left(M_{S_1}^2-M_{S_2}^2\right)}\\ &-\frac{2h\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\left(M_{S_1}^2-M_{S_2}^2\right)}\\ &+\frac{2h\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\left(M_{S_1}^2-M_{S_2}^2\right)}\\ &+hC_{H2}H_{a_2}H_{a_1}^\dagger X_{H1,a_1,a_2}\lambda_{H1,a_1,a_2} \end{aligned}$	$\begin{aligned} &-\frac{2\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon}\\ &-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \end{aligned}$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$F B_{\mu\nu}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$ $-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)B_{\mu_1\mu_2}^2$	$+\frac{3\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{2\epsilon}$ $+\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{\hbar H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H2,a_1,a_3}\mathcal{I}_{H2,a_2,a_4}(\lambda_{H2})^2}{2\epsilon}$ $+\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)H_{a_3}H_{a_4}H_{a_1}^\dagger H_{a_2}^\dagger\mathcal{I}_{H2,a_1,a_3}\mathcal{I}_{H2,a_2,a_4}(\lambda_{H2})^2$ $+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3\epsilon}}$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S1}^2-M_{S2}^2)}$ $+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S1}^2-M_{S2}^2)}$ $-\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S1}^2-M_{S2}^2)}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}(M_{S1}^2-M_{S2}^2)}$ $+CH21\hbar H_{a_2}H_{a_1}^\dagger\mathcal{I}_{H2,a_1,a_2}$	$-\frac{2\hbar\left(W_{\mu_1\mu_2}^{A1}\right)^2}{3\epsilon}$ $-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\left(W_{\mu_1\mu_2}^{A1}\right)^2$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$ $-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)B_{\mu_1\mu_2}^2$	$+\frac{3\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2}{2\epsilon}$ $+\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H1})^2$ $+\frac{3\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2}{2\epsilon}$ $+\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger(\lambda_{H2})^2$ $+C_{H2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2}$ $+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2}$ $-\frac{2\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2}$	$-\frac{5\hbar\left(W_{\mu_1\mu_2}^{A1}\right)^2}{6\epsilon}$ $-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\left(W_{\mu_1\mu_2}^{A1}\right)^2$ $-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\left(W_{\mu_1\mu_2}^{A1}\right)^2$	$-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H$	$F\ W_{\mu\nu}$	$H\ e$	$F\ H\ e$
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